

Question 1 (Switch):

```
1. main()
2. {
3. char chr;
4. printf("Enter the special character\n");
5. scanf ("%c", &chr);
6. if (chr != 48) && (chr != 49) && (chr != 50) && (chr != 51) && (chr != 52) &&
   (chr != 53) && (chr != 54) && (chr != 55) && (chr != 56) && (chr != 57)
7. {
8. switch(chr)
9. {
10. Case '*': printf("It is a special character");
11. break;
12. Case '#': printf("It is a special character");
13. break;
14. Case '@': printf("It is a special character");
15. break;
16. Case '!': printf("It is a special character");
17. break;
18. Case '%': printf("It is a special character");
19. break;
20. default : printf("You have not entered a special character");
21. break;
22. } // end of switch
23. } // end of If
24. else
25. printf("You have not entered a character");
26. } // end of main()
```

(a) Draw the Control Flow graph for the program.

(b) Calculate the cyclomatic complexity.

(c) List all independent paths.

(d) Design test cases from independent paths

Question 2:

```
1. main()
2. {
3. int number; int fact();
4. clrscr();
5. printf("Enter the number whose factorial is to be found out");
6. scanf("%d", &number);
7. if (number < 0)
8.
printf("Facorial cannot be defined for this number);
9. else
10.
printf("Factorial is %d", fact(number));
11. }
12. int fact(int number)
13. {
14. int index;
15. int product = 1;
16. for(index=1; index<=number; index++)
17.
product = product * index;
18. return(product);
19. }
```

(a) Draw the Control Flow graph for the program.

(b) Calculate the cyclomatic complexity of the program.

(c) List all independent paths.

(d) Design test cases from independent paths

Question 3 (Recursion):

```
1. #include<stdio.h>
2. long int multiplyNumbers(int n);
3. int main() {
4.     int n;
5.     printf("Enter a positive integer: ");
6.     scanf("%d",&n);
7.     printf("Factorial of %d = %ld", n, multiplyNumbers(n));
8.     return 0;
9. }

10. long int multiplyNumbers(int n) {
11.     if (n>=1)
12.         return n*multiplyNumbers(n-1);
13.     else
14.         return 1;
15. }
```

- (a) Draw the Control Flow graph for the program.
- (b) Calculate the cyclomatic complexity of the program.
- (c) List all independent paths.
- (d) Design test cases from independent paths

Question 4 (while loop):

```
1. #include <stdio.h>
2. int main() {
3.     int num, count = 0;
4.     printf("Enter a number: ");
5.     scanf("%d", &num);
6.     if (num == 0)
7.         count = 1;
8.     else {
9.         while (num != 0) {
10.            num = num / 10;
11.            count++;
12.        }
13.    }
14.    printf("Total digits = %d\n", count);
15.    return 0;
16. }
```

- (a) Draw the Control Flow graph for the program.
- (b) Calculate the cyclomatic complexity of the program.
- (c) List all independent paths.
- (d) Design test cases from independent paths

Question 5 (for loop):

```
1. #include <stdio.h>
2. int main() {
3. char string[80];
4. int index;
5. printf("Enter the string for checking its characters: ");
6. scanf("%s", string);
7. for (index = 0; string[index] != '\0'; ++index) {
8. if (string[index] >= '0' && string[index] <= '9')
9. printf("%c is a digit\n", string[index]);
10. else if ((string[index] >= 'A' && string[index] <= 'Z') || (string[index] >=
'a' && string[index] <= 'z'))
11. printf("%c is an alphabet\n", string[index]);
12. else
13. printf("%c is a special character\n", string[index]);
14. }
15. return 0;
16. }
```

(a) Draw the Control Flow graph for the program.

(b) Calculate the cyclomatic complexity of the program.

(c) List all independent paths.

(d) Design test cases from independent paths

Question 6 (for loop):

```
1. #include <stdio.h>
2. int main() {
3.     int num;
4.     printf("Enter a number: ");
5.     scanf("%d", &num);
6.     if (num > 0) {
7.         if (num % 2 == 0)
8.             printf("%d is a positive even number\n", num);
9.         else
10.            printf("%d is a positive odd number\n", num);
11.     }
12.     else if (num < 0) {
13.         if (num % 2 == 0)
14.             printf("%d is a negative even number\n", num);
15.         else
16.             printf("%d is a negative odd number\n", num);
17.     }
18.     else {
19.         printf("The number is zero\n");
20.     }
21.     return 0;
22. }
```

(a) Draw the Control Flow graph for the program.

(b) Calculate the cyclomatic complexity of the program.

(c) List all independent paths.

(d) Design test cases from independent paths